



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Pizza Sales Analysis Dashboard using Power BI

The domain of the Project:

SQL & PowerBI

Team Mentors (and their designation):

SRAVAN NEMANA(Co-Founder of stealth AI Startup)

Team Members:

Mr. Samir Kamble

Period of the project

July 2025 to December 2025



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Declaration

The project titled “**Pizza Sales Analysis Dashboard**” has been mentored by **SRAVAN NEMANA** Sir, organised by SURE Trust, from July 2025 to December 2025, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Members:

Mr. Samir Kamble -- Signature

Mentor's Name - SRAVAN NEMANA

Designation— Co-Founder of stealth AI Startup

Prof. Radhakumari

Executive Director & Founder

SURE Trust



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Executive Summary

The Pizza Sales Analysis Dashboard using Power BI project focuses on analyzing a pizza sales dataset to derive meaningful business insights and support data-driven decision-making in the food and restaurant industry. The dataset contains detailed information related to pizza orders, including pizza types, categories, sizes, quantities, pricing, and order timestamps. The primary objective of the project is to analyze sales performance, understand customer preferences, and identify key trends through interactive visualizations.

The project follows a structured data analytics approach, beginning with data cleaning and transformation using Power Query in Power BI. Data types were validated, inconsistencies were addressed, and a clear data structure was established to ensure accuracy and reliability. A data dictionary was created to understand each attribute clearly. After data preparation, Key Performance Indicators (KPIs) such as Total Revenue, Total Pizzas Sold, Average Order Value, Best-Selling Pizza, and Peak Sales Hour/Day were defined and calculated using DAX.

An interactive Power BI dashboard was developed to present these KPIs in a clear and visually appealing manner. The dashboard includes KPI cards, charts, and filters that allow users to explore sales trends by category, size, and time period. A Pizza Hut–style theme was applied to enhance usability and align the dashboard with a real-world restaurant business context.

The analysis revealed important insights, including top-selling pizza varieties, high-revenue categories, and peak ordering times. These insights can help businesses improve inventory planning, staffing decisions, and promotional strategies. Overall, this project demonstrates how business intelligence tools like Power BI can transform raw sales data into actionable insights, making it valuable for executives and stakeholders seeking to improve operational efficiency and business performance.



Introduction

Background and Context of the Project

The food and restaurant industry generates a significant amount of transactional data on a daily basis, including customer orders, product details, pricing, and time-based sales information. In recent years, the ability to analyze this data effectively has become a critical factor for business success. Restaurants that leverage data analytics can better understand customer preferences, optimize operations, and increase profitability. However, many small and medium-sized food businesses still rely on manual reporting or basic summaries, which limits their ability to gain deeper insights.

This project, “Pizza Sales Analysis Dashboard using Power BI,” is designed to address this gap by applying data analytics techniques to a real-world pizza sales dataset. The project uses Power BI, a powerful business intelligence and data visualization tool, to transform raw sales data into meaningful and interactive insights. By analyzing historical sales data, the project aims to provide a clear understanding of revenue patterns, product performance, and customer purchasing behavior.

Problem Statement and Goals of the Project

Despite having access to detailed sales data, businesses often struggle to derive actionable insights due to the lack of structured analysis and visualization. Raw datasets containing thousands of records can be difficult to interpret without proper tools and methodologies. Important business questions such as Which pizzas generate the highest revenue?, When do customers place the most orders?, and What is the average value of an order? often remain unanswered or are answered inaccurately.

The primary goal of this project is to analyze pizza sales data systematically and present insights in a clear and understandable manner. The specific goals of the project include:

- Identifying key sales performance indicators such as total revenue and total pizzas sold
- Understanding customer preferences based on pizza categories and sizes
- Analyzing order patterns to determine peak sales hours and days
- Building an interactive dashboard that supports data-driven decision-making

Scope and Limitations of the Project

The scope of this project is limited to the analysis of historical pizza sales data provided in the dataset. The analysis focuses on sales-related metrics such as revenue, quantity sold, order value, and time-based trends. The project emphasizes descriptive analytics, which helps in understanding what has happened in the past and identifying patterns within the data.

However, the project also has certain limitations. The dataset does not include customer demographic information such as age, location, or purchasing history, which limits deeper customer segmentation analysis. Additionally, the project does not incorporate predictive analytics or real-



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time data streaming. The insights derived from the dashboard are dependent on the accuracy and completeness of the provided dataset.

Innovation Component in the Project

The innovation in this project lies in the transformation of static sales data into a dynamic and interactive decision-support system. Instead of using traditional reports or spreadsheets, the project leverages Power BI's advanced visualization capabilities to create an intuitive dashboard. The use of DAX measures for KPI calculation and Power Query for data transformation adds analytical depth to the project.

Another innovative aspect is the application of a restaurant-inspired (Pizza Hut-style) dashboard theme, which enhances user experience and makes the analysis more relatable to real-world business scenarios. The project also demonstrates how business intelligence tools can be used by non-technical stakeholders to explore data interactively, making analytics more accessible and impactful.



Project Objectives

The primary objective of this project is to analyze pizza sales data and convert raw transactional information into meaningful business insights using data analytics and visualization techniques. The project aims to support data-driven decision-making in the food and restaurant industry by presenting key performance indicators through an interactive dashboard.

Specific Objectives of the Project

1. To analyze the overall sales performance of pizza orders by calculating total revenue generated over the given period.
2. To determine the total quantity of pizzas sold and understand customer demand trends.
3. To calculate the average order value in order to assess customer spending behavior.
4. To identify the best-selling pizza varieties based on sales quantity and revenue contribution.
5. To analyze time-based sales patterns and identify peak sales hours and days.
6. To design an interactive Power BI dashboard that allows users to explore sales data dynamically using filters and slicers.

Expected Outcomes and Deliverables

- A cleaned and well-structured dataset prepared using Power Query.
- A set of well-defined KPIs created using DAX measures for accurate analysis.
- An interactive Power BI dashboard displaying key insights related to revenue, sales volume, customer preferences, and peak sales periods.
- Visual representations such as KPI cards, charts, and trend analysis to support quick decision-making.
- A project report documenting objectives, methodology, results, and conclusions.



Methodology and Results

Methods and Technology Used

This project follows a structured data analytics and business intelligence methodology to analyze pizza sales data and extract meaningful insights. The approach begins with understanding the dataset and identifying business requirements, followed by data cleaning, transformation, analysis, and visualization. The project primarily uses descriptive analytics, which focuses on analyzing historical data to identify patterns, trends, and performance metrics.

Key analytical techniques used in this project include:

- Data preprocessing and transformation
- Aggregation and summarization of sales data
- KPI-based analysis using calculated measures
- Time-based trend analysis
- Interactive data visualization

The project emphasizes converting raw transactional data into a decision-support system that can be easily interpreted by business stakeholders.

Tools and Software Used

The following tools and software were used to complete the project:

- Power BI Desktop – Used for data modeling, KPI creation, and dashboard development
 - Power Query – Used for data cleaning, transformation, and preparation
 - DAX (Data Analysis Expressions) – Used to create calculated measures and KPIs
 - Microsoft Excel / CSV File – Used as the source format for the dataset
 - GitHub – Used for project version control and documentation
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Data Collection Approach

The dataset used in this project is a historical pizza sales dataset obtained from a publicly available source. The data represents transactional records of pizza orders placed over a specific time period. Since the dataset is pre-collected, no real-time or primary data collection methods such as surveys or interviews were used. The focus of the project is on analyzing existing data to generate insights rather than collecting new data.

Project Architecture

The architecture of the project follows a clear and logical flow:

1. Data Source Layer



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- Pizza sales data stored in CSV/Excel format

2. Data Processing Layer

- Data imported into Power BI
- Data cleaning and transformation using Power Query
- Validation of data types and removal of inconsistencies

3. Analytics Layer

- Creation of calculated columns and DAX measures
- Definition of KPIs such as Total Revenue, Total Pizzas Sold, Average Order Value, Best-Selling Pizza, and Peak Sales Hour

4. Visualization Layer

- Development of interactive dashboards
- Use of KPI cards, bar charts, line charts, and slicers
- Application of a Pizza Hut–style theme for professional presentation

5. Presentation Layer

- Insights delivered through a user-friendly Power BI dashboard
- Results documented in the project report and GitHub repository

This layered architecture ensures clarity, scalability, and ease of understanding.

Results and Dashboard Explanation

The final Power BI dashboard provides a comprehensive view of pizza sales performance. KPI cards display high-level metrics such as total revenue, total pizzas sold, and average order value. Visual charts highlight the best-selling pizza varieties and category-wise sales performance. Time-based visuals reveal peak sales hours and days, enabling a deeper understanding of customer ordering behavior.

Interactive slicers allow users to filter data by pizza category, size, and time period, making the dashboard flexible and user-friendly. The use of consistent color themes and clear labeling improves readability and enhances the overall user experience.

(Final dashboard screenshots should be attached here, showing KPI cards, charts, and slicers with brief captions explaining each visual.)

Project GitHub Link



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The complete project, including the dataset, Power BI file, and documentation, is available on GitHub:

GitHub Repository: <https://github.com/samkamble523/Pizza-Sales-Dashboard>



Learning and Reflection

New Learnings

Through this project, I gained practical knowledge and hands-on experience in data analytics and business intelligence using Power BI. I learned how to transform raw sales data into meaningful insights by following a structured analytical approach. I developed skills in Power Query for data cleaning and transformation, including handling missing values, correcting data types, and preparing data for analysis.

I also learned to use DAX to create key performance indicators such as Total Revenue, Total Pizzas Sold, Average Order Value, and Peak Sales Time. This improved my analytical thinking and understanding of how business metrics are calculated and interpreted. In addition, I learned the importance of defining clear project objectives, selecting relevant KPIs, and designing dashboards that effectively communicate insights to stakeholders.

Overall Experience

Working on this project as an individual was a valuable learning experience. I was involved in all stages of the project, from data understanding and analysis to dashboard design and documentation. Initially, learning Power BI and DAX was challenging, but consistent practice helped me gain confidence.

Designing an interactive and visually appealing dashboard was especially rewarding, as it demonstrated how data analytics can support real-world business decision-making. Overall, this project enhanced my technical, analytical, and problem-solving skills and strengthened my interest in pursuing further work in data analytics and business intelligence.



Conclusion and Future Scope

Conclusion

The Pizza Sales Analysis Dashboard using Power BI successfully achieved its intended objectives of analyzing sales data and presenting meaningful business insights through interactive visualizations. The project effectively transformed raw pizza sales data into structured information by applying data cleaning, transformation, and analytical techniques. Key performance indicators such as Total Revenue, Total Pizzas Sold, Average Order Value, Best-Selling Pizza, and Peak Sales Hour were accurately calculated and visualized using Power BI.

The interactive dashboard provides a clear overview of sales performance, customer preferences, and time-based ordering patterns. The insights derived from the analysis can help restaurant managers make informed decisions related to inventory management, staffing, and promotional strategies. Overall, the project demonstrates the practical application of data analytics and business intelligence tools in solving real-world business problems.

Future Scope

The scope of this project can be expanded in several ways. Future enhancements may include integrating customer demographic and feedback data to enable deeper customer segmentation and sentiment analysis. Predictive analytics and sales forecasting models can be implemented to estimate future demand and support proactive decision-making.

Additionally, the dashboard can be connected to real-time data sources to enable live monitoring of sales performance. Advanced visualizations and machine learning techniques can further improve insights. These enhancements would make the system more robust, scalable, and valuable for business stakeholders.